

An Internship Report
Of
C/C++ Programming Internship at
Cognifyz IT Solutions Pvt. Ltd.

Submitted
To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of
B.Tech (Computer Science and Engineering)



By

Roll Number of Student: 2410030988

Name of Student: ANURODH SINGH

Batch: 2024-2028

CANDIDATE'S DECLARATION

I, ANURODH SINGH, do hereby declare that the internship report titled "C/C++ Programming Internship at Cognifyz IT Solutions Pvt. Ltd." has been prepared to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All references have been acknowledged, and I have not presented material from any other source as my own. I affirm that the report is aligned to the internship domain stated on the supplied completion certificate and has not been submitted to any other institute for any degree or diploma requirement. I shall be solely responsible for any copyright violation in this regard.

Signature:

Student Name: ANURODH SINGH

Roll No.: 2410030988

ACKNOWLEDGEMENT

I take this opportunity to express my gratitude to everyone who supported me throughout the internship program. Their guidance, constructive feedback, and encouragement helped me strengthen my understanding of programming logic and practical problem solving.

I sincerely thank Cognifyz IT Solutions Pvt. Ltd. for providing me the opportunity to work as a C/C++ Programming Intern. The internship domain encouraged structured thinking, careful implementation, debugging, and continuous learning. The certificate specifically recognizes dedication, sincerity, coordination skills, effective communication, attention to detail, enthusiasm, commitment, and a strong desire to learn.

I am also thankful to IILM University, Greater Noida, and the faculty of the School of Computer Science and Engineering for their continuous support and encouragement. Finally, I express my gratitude to my parents and well-wishers for their blessings.

Signature:

Student Name: ANURODH SINGH

Roll No.: 2410030988

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3. INTERNSHIP COMPLETION CERTIFICATE



Cognifyz IT Solutions

Internship Completion Certificate

Date -11/05/2026

This is to certify that **Anurodh Singh (Intern ID: CTI/A1/C337566)**, currently pursuing B.Tech from The IILM UNIVERSITY, was working as a **C/C++ Programming Intern** with Cognifyz IT Solutions Pvt. Ltd. from 11/04/2026 to 11/05/2026.

During this period, he has served as a C/C++ Programming Intern and has displayed remarkable dedication, sincerity, and a strong desire to learn. He has exhibited exceptional coordination skills and effective communication abilities. Moreover, his attention to detail has been truly impressive.

He has consistently approached new assignments and challenges with enthusiasm, showcasing His passion for C/C++ Programming. His commitment and willingness to acquire new knowledge and skills have been evident throughout His internship.

We extend our best wishes to Anurodh Singh for a successful future, and we have no doubt that he will continue to excel in the field of C/C++ Programming.

With Regards,
Cognifyz IT Solutions Pvt. Ltd.



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INTERNSHIP DETAILS

Student Name:	Anurodh Singh
Intern ID:	CTI/A1/C337566
Internship Role:	C/C++ Programming Intern
Organization:	Cognifyz IT Solutions Pvt. Ltd.
Duration:	11 April 2026 - 11 May 2026
Certificate Date:	11 May 2026
University:	IILM University

This page summarizes the factual details visible on the supplied internship completion certificate. The certificate does not include a detailed task sheet or offer letter; therefore, the technical sections that follow are organized around the verified C/C++ Programming internship domain.

4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the internship domain stated on the completion certificate issued to Anurodh Singh: C/C++ Programming Intern at Cognifyz IT Solutions Pvt. Ltd. from 11 April 2026 to 11 May 2026. The internship was undertaken while pursuing B.Tech at IILM University. Its central technical theme is C/C++ programming, which requires translating a problem into an algorithm, implementing the logic correctly, compiling the source code, testing outputs, and refining the solution.

Because the supplied certificate confirms the role and duration but does not provide a detailed task log, the technical sections of this report organize representative C/C++ concepts that are directly aligned with the verified internship domain. The certificate-verified professional qualities are kept separate from the domain-aligned technical content.

4.2 Organization Profile

Cognifyz IT Solutions Pvt. Ltd. is identified on the supplied certificate with the Cognifyz brand and the tagline "Where Data Meets Intelligence". The certificate confirms the internship role as C/C++ Programming Intern under Intern ID CTI/A1/C337566 and recognizes the intern for dedication, sincerity, communication, coordination, attention to detail, enthusiasm, and willingness to acquire new knowledge and skills.

4.3 Problem Statement

C and C++ programming require more than memorizing syntax. A programming problem must be understood, decomposed into steps, expressed using appropriate data types and control structures, implemented in modular code, compiled without errors, and tested for correct results. In C++, object-oriented concepts and standard library components further help organize larger solutions. The challenge is therefore to build correct, readable, maintainable programs while developing disciplined debugging and problem-solving habits.

4.4 Project Objectives

- To understand the structure, syntax, compilation, and execution of C and C++ programs.
- To strengthen logic building using conditions, loops, functions, arrays, and strings.
- To understand pointers, structures, memory-oriented concepts, and file handling in C.
- To study classes, objects, encapsulation, inheritance, and polymorphism in C++.
- To become familiar with standard library containers and common searching/sorting ideas.
- To practise debugging by identifying syntax, runtime, and logic errors.
- To verify solutions using dry runs, sample inputs, and boundary test cases.
- To communicate program logic, output, and learning clearly and systematically.

4.5 Scope of the Project

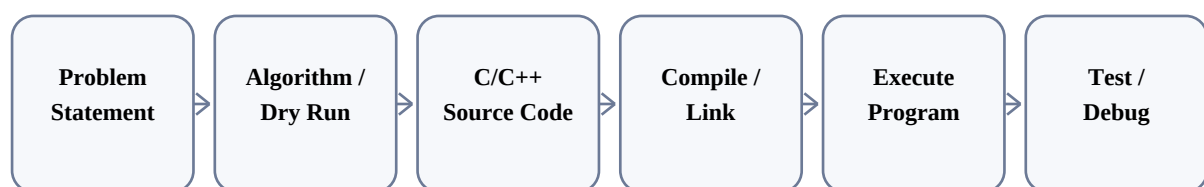
The report covers programming fundamentals, control flow, functions, arrays and strings, pointers and structures, basic file handling, object-oriented programming in C++, selected standard-library concepts, common data structures, debugging, testing, and console-application workflow. The scope is educational and role-aligned. The completion certificate itself does not enumerate specific coding assignments, repositories, or project deliverables; therefore, no unverified project result is presented as certificate-confirmed evidence.

4.6 Technologies and Tools Used

- Programming languages: C and C++.
- Compiler workflow: source code -> compiler/linker -> executable -> console output.
- Programming concepts: input/output, operators, conditions, loops, functions, arrays, strings, pointers, structures, classes and objects.
- C++ library concepts: string, vector, map/set and algorithms such as sort/search where appropriate.
- File handling: text-based read/write operations for simple record-oriented programs.
- Debugging and testing: compiler diagnostics, dry runs, sample cases, and edge-case checking.
- Development environment: any standard C/C++ IDE/editor and compiler suitable for the above workflow.

4.7 System Architecture

The programming workflow follows a layered execution path. Each stage transforms the problem into a more concrete form and provides a checkpoint for verification:



The architecture emphasizes correctness at every stage: a clear algorithm reduces coding mistakes, compiler feedback catches syntax/type issues, and test cases expose incorrect conditions or boundary handling.

4.8 Methodology and Task Execution

The internship certificate verifies the C/C++ programming role but does not list individual task sheets. Accordingly, the methodology below is a domain-aligned structure used to organize the technical report without presenting unverified assignments as certificate-confirmed work.

Module	Technical Area	Domain-Aligned Focus
L1-T1	C Fundamentals	Program structure, data types, operators, console input/output and compilation.
L1-T2	Control Flow & Functions	if/switch, loops, functions, arrays and strings for modular logic.
L1-T3	Pointers, Structures & Files	Memory-oriented concepts, record structures and basic persistent file I/O.
L2-T1	C++ OOP	Classes, objects, encapsulation, inheritance and polymorphism.
L2-T2	STL & Data Structures	Standard containers plus stack, queue, linked-list and searching/sorting concepts.
L2-T3	Debugging & Testing	Compiler diagnostics, dry runs, sample cases, boundary cases and output verification.

The technical learning path moves from syntax and control flow to modular programming, object-oriented design, data structures, debugging, and reliable output verification.

4.8.1 Level 1 Learning: C Fundamentals, Functions, and Memory

C provides a direct view of program structure and control flow. A typical learning sequence starts with variables, constants, operators, formatted input/output, decision statements, and loops. These concepts form the base for expressing algorithms in executable form.

Functions improve modularity by separating a larger solution into smaller reusable units. Arrays and strings provide structured ways to process collections of values and text, while pointer concepts introduce address-based access and help explain how arrays, function arguments, and dynamic memory are related.

Structures group related fields into records, and file handling extends a program beyond temporary console input by allowing simple records to be written and read from storage. These topics encourage careful attention to data types, indexing, input validation, and resource handling.



Figure 1. Domain-aligned C programming structure: input, control flow, modular logic, and output.

4.8.2 Level 2 Learning: C++ OOP, Standard Library, and Debugging

C++ builds on procedural programming and adds object-oriented mechanisms. Classes and objects combine data with functions, encapsulation controls access, inheritance supports code reuse, and polymorphism allows a common interface to represent different behaviors.

The C++ standard library provides ready-made types and algorithms. Containers such as vector, string, map, and set can simplify common storage and lookup tasks, while standard algorithms reduce repetitive code for operations such as sorting and searching.

Debugging remains essential even when a program compiles. Syntax errors are typically reported by the compiler; runtime problems may appear only during execution; and logic errors can produce plausible but incorrect outputs. Dry runs and test cases are therefore part of the programming process rather than an afterthought.

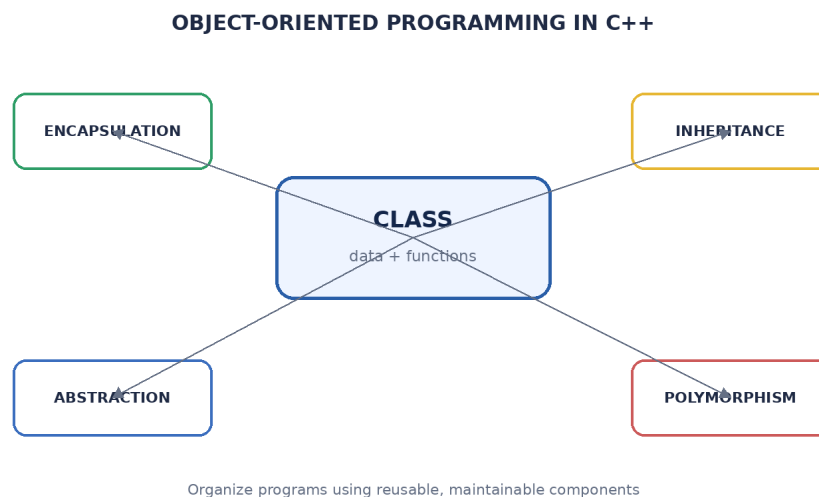


Figure 2. Object-oriented programming concepts used to organize C++ solutions.

4.8.3 Programming Practice and Code Quality

Readable code improves correctness, debugging, and communication. The following practices summarize a disciplined C/C++ programming approach:

Practice	Purpose	Programming Benefit
Descriptive names	Make variables/functions self-explanatory	Improves readability
Functions	Separate repeated or independent logic	Supports modularity
Input validation	Reject or handle invalid values	Prevents avoidable errors
Dry run	Trace values before/after coding	Finds logic mistakes
Test cases	Check normal and boundary inputs	Improves reliability
Comments / notes	Explain non-obvious logic and output	Supports communication

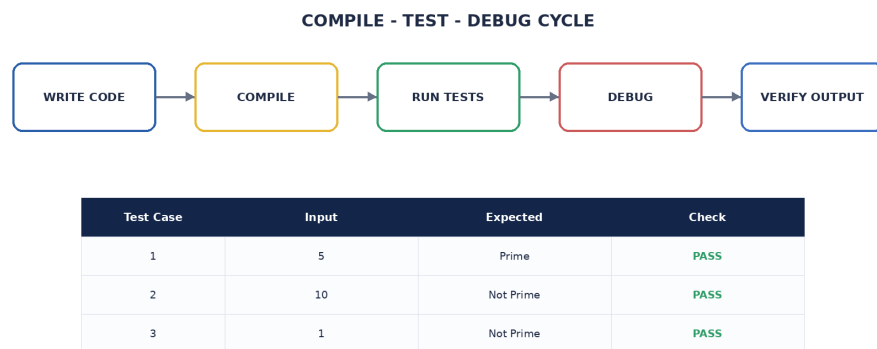


Figure 3. Compile-test-debug cycle used to verify programming logic.

4.9 Outcomes and Key Learning

The supplied certificate records strong dedication, sincerity, a desire to learn, coordination skills, effective communication, attention to detail, enthusiasm, commitment, and willingness to acquire knowledge and skills. Within the verified C/C++ Programming domain, these qualities are especially relevant to careful coding, debugging, test-case checking, and clear explanation of program logic. The technical sections of this report therefore focus on programming fundamentals and disciplined problem solving while the professional claims remain limited to what is stated on the certificate.

4.10 Certificates of Completion and Communication Proof

The Internship Completion Certificate supplied with this report confirms Anurodh Singh as a C/C++ Programming Intern at Cognifyz IT Solutions Pvt. Ltd. under Intern ID CTI/A1/C337566 for the period 11 April 2026 to 11 May 2026. The certificate is dated 11 May 2026.

The certificate also records professional observations about dedication, sincerity, coordination, effective communication, attention to detail, enthusiasm, commitment, and willingness to acquire new knowledge and skills.

Evidence	Verified Detail
Completion Certificate	Official certificate image supplied by the user.
Student / Intern	Anurodh Singh - Intern ID CTI/A1/C337566.
Role	C/C++ Programming Intern.
Organization	Cognifyz IT Solutions Pvt. Ltd.
Duration	11/04/2026 to 11/05/2026.
Professional Remarks	Dedication, sincerity, desire to learn, coordination, communication and attention to detail.

No separate offer letter, coding repository, task log, or assignment sheet for Anurodh Singh was supplied in the current materials. For that reason, the technical modules in this report are explicitly domain-aligned content rather than claimed certificate-verified task evidence.

5. BIBLIOGRAPHY/REFERENCES

1. Cognifyz IT Solutions Pvt. Ltd., "Internship Completion Certificate", Intern ID CTI/A1/C337566, dated 11 May 2026.
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6. GCC Documentation, GNU Compiler Collection - C and C++ compilation workflow.
7. cppreference, reference material for C and C++ language/library concepts.